Task 3: Research Results, Data Analysis and Scientific Visualization

The Effectiveness of AI-Based Adaptive Platforms on Student Academic Performance in Higher Education

Nurali Seifulla & Nurbakyt Sagidulla · Narxoz University

November 2026

1. Submission links

Item	Location
GitHub repository	https://github.com/nurba-git/research-methods (release tag `v1.0-task3`)
Analysis notebook	`notebooks/01_data_analysis.ipynb` (executed, outputs saved)
Processed metrics	`data/processed/metrics_summary.csv`
Figures	`reports/figures/fig1–fig3` as `.pdf` and 300-dpi `.png`
Domain model	`docs/domain_model.md` (ML track: LaTeX loss and metrics)
Methodology card	`docs/methodology_card.pdf` (revised after Task 2 feedback)

Data status. The real Week 7–8 platform export is not yet available. All results below come from a **simulated** log (`src/simulate_data.py`, seed 42) written in the exact planned export format. They show that the pipeline works; they are not empirical findings. Replacing `data/raw/test_run.jsonl` with the real export and re-running the same commands regenerates every number and figure.

2. Changes made after the Task 2 feedback

Feedback	Fix
Authors given by first names only	Full names (Nurali Seifulla, Nurbakyt Sagidulla) in card, README, LICENSE, citation
Design left open (RCT or quasi-experiment)	Fixed: two-arm parallel-group RCT, 1:1, stratified by pre-test tercile, N = 140 from a power analysis
PDF showed raw Markdown	Card rendered with pandoc/xelatex: `docs/methodology_card.pdf`
Project nested in a subfolder; README commands fail from root	Project moved to repository root; quickstart tested from a fresh clone
No README on the GitHub landing page	`README.md` now at the root
Uninformative commit messages	13 new commits with descriptive Conventional-Commit messages and bodies

3. Pipeline

1. **Ingestion and warm-up filtering.** 1 374 JSONL events (pre-test, 8 weekly sessions, post-test). Sessions 1–2 are onboarding and cold-start calibration of the adaptive model; the 280 warm-up sessions are kept in the raw file but excluded from all session-level statistics.
2. **Validity and guardrails.** Out-of-range scores, non-positive times or latencies stop the run. Error rate and attrition are reported with Wilson 95% CIs; p95 latency is checked against 1000 ms.
3. **Aggregation.** Per metric and arm: mean, SD, median, p25/p75, p95, p99, and percentile-bootstrap 95% CIs of the mean and median (10 000 resamples, seed 42).
4. **Hypothesis test.** Welch t-test on score gain plus bootstrap CI of the difference, Hedges' g; sensitivity analyses: ANCOVA `posttest ~ arm + pretest` and Mann–Whitney U.
5. **Visualization.** `src/visualize.py`; colour-blind palette plus markers, line styles and hatching.

The logic lives in `src/process_data.py` and `src/visualize.py`; the notebook imports these modules, so the notebook and the command-line run produce identical files.

4. Results (simulated data)

Students analysed (complete cases): 133 — traditional 66, adaptive 67.

Metric (unit)	Traditional (baseline)	AI-adaptive (proposed)	Δ, 95% CI
Score gain (points), mean ± SD	12.2 ± 7.3	15.2 ± 7.1	**+2.95 [0.54, 5.41]**
Score gain, median (p25–p75)	12.0 (7.0–16.0)	15.0 (10.0–19.0)	–
Pre-test (points), mean ± SD	54.2 ± 9.3	54.4 ± 8.7	balance
Time on task (min/session), mean ± SD	62.7 ± 4.3	54.4 ± 4.7	−8.3
Quiz score (%), median (p25–p75)	70.0 (61.2–78.9)	74.0 (64.8–82.6)	–
Latency (ms), median (p25–p75)	186 (131–246)	319 (233–422)	–
Latency (ms), p95 / p99	399 / 466	621 / 871	≤ 1000 (pass)
Error rate, evaluation (Wilson CI)	0.73% [0.25, 2.12]	1.22% [0.52, 2.83]	≤ 2% (pass)
Attrition (Wilson CI)	7.0% [3.1, 15.5]	2.9% [0.8, 10.0]	≤ 15% (pass)

Test. Welch $t(130.8) = 2.36$, $p = 0.020$; Hedges' $g = 0.41$; Mann–Whitney $p = 0.018$. ANCOVA adjusted effect +2.93 points, 95% CI [0.47, 5.40], $p = 0.020$. Under the pre-registered decision rule H_0 is rejected for this simulated dataset.

5. Discussion

- The simulated adaptive arm gains about 3 points more than the baseline ($g \approx 0.4$). Pre-test means are balanced, so unadjusted and ANCOVA estimates agree, as expected under randomization.
- Warm-up filtering matters: in sessions 1–2 the adaptive arm scores below the baseline and its latency is about 1 s (cold start). Keeping those sessions would bias the comparison against it.
- All guardrails pass on point estimates. The upper Wilson bound of the adaptive error rate (2.8%) is above 2%, so this guardrail needs monitoring on real data.
- Limitations: simulated data, single course, no blinding of students, complete-case analysis.

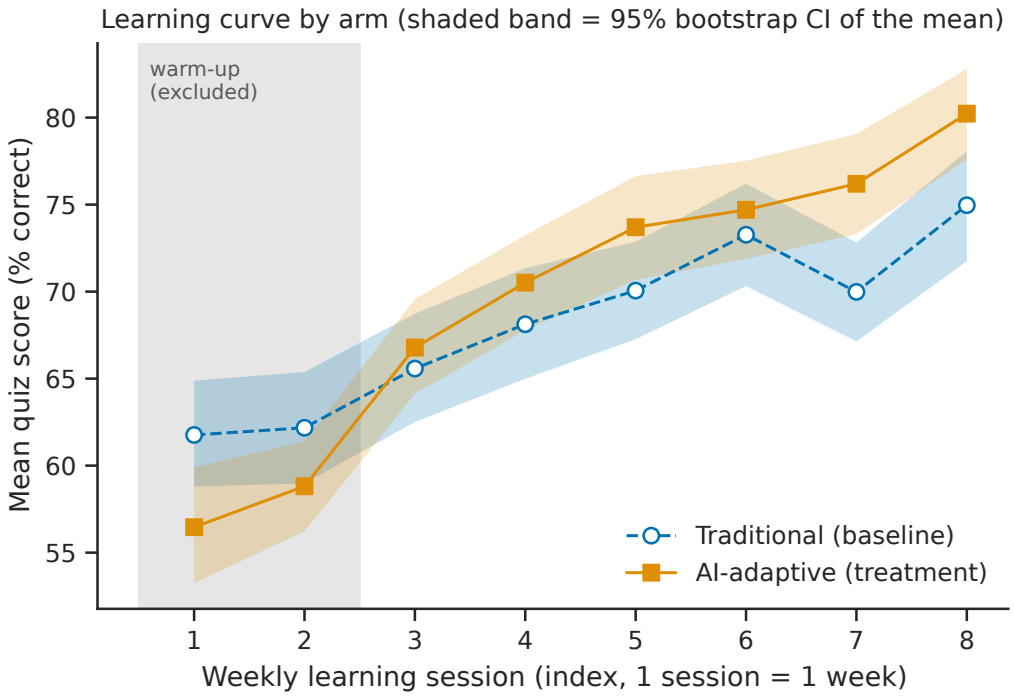


Figure 1: Learning curve: mean weekly quiz score with 95% bootstrap CI bands. Warm-up sessions are shaded.

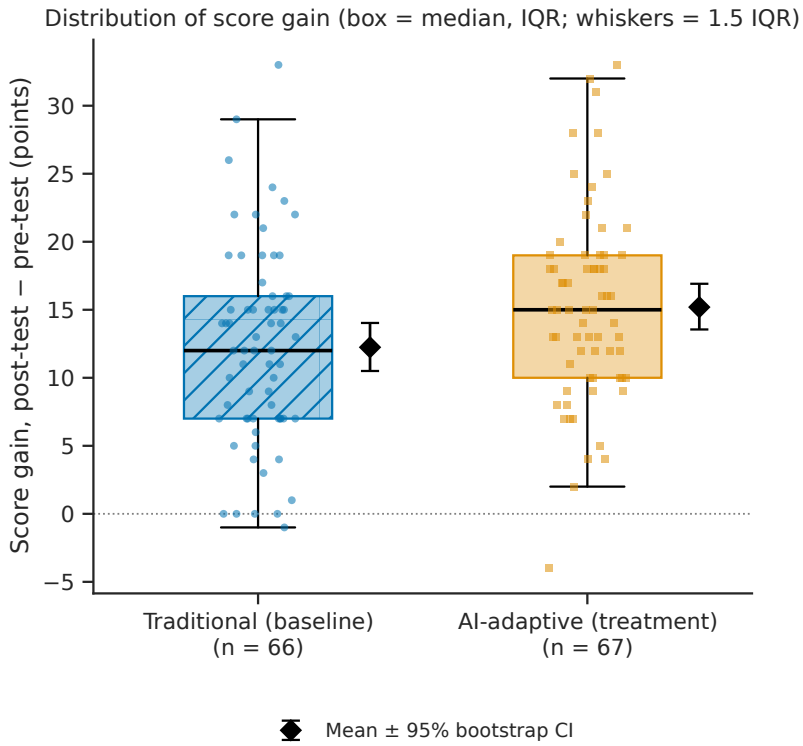


Figure 2: Distribution of score gain per arm with mean and 95% bootstrap CI.

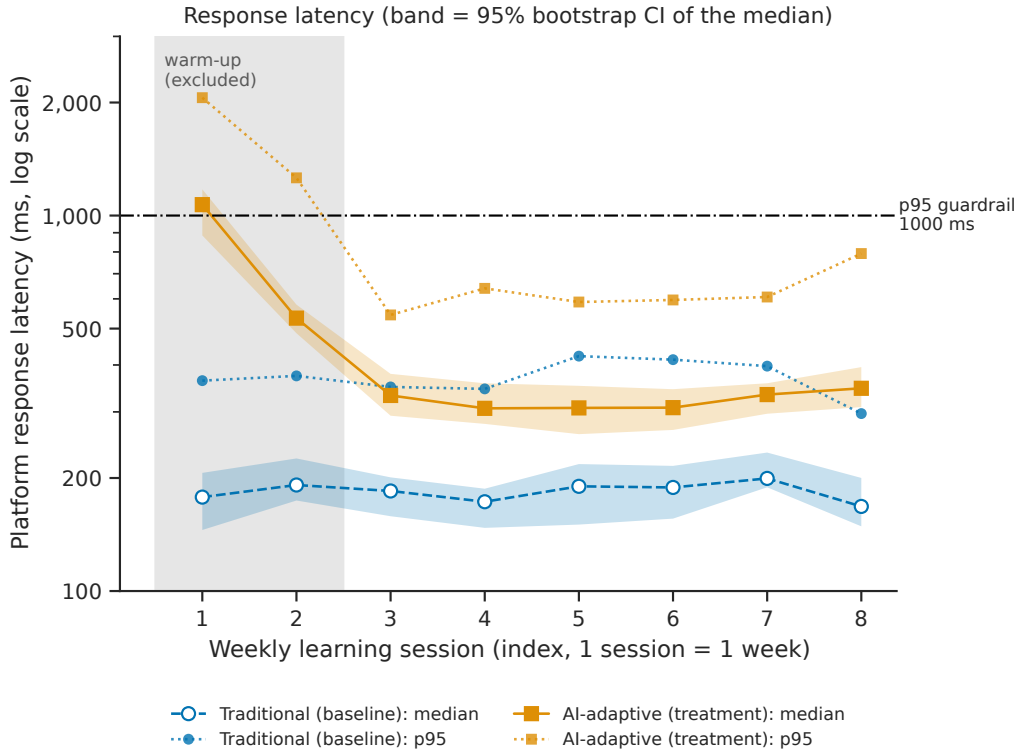


Figure 3: Platform response latency (log scale): median with 95% CI band and p95 against the guardrail.

6. Domain model (ML track) — key definitions

Learner model and loss (full notation table and system diagram in `docs/domain_model.md`):

$$\hat{p}_{ijt} = \sigma\Big(\theta_i - \beta_j + \sum_k q_{jk}(\gamma_k s_{ikt} + \rho_k f_{ikt})\Big)$$

$$\mathcal{L}(\mathbf{w}) = -\frac{1}{|\mathcal{D}|}\sum_{(i,j,t)\in\mathcal{D}} [y_{ijt}\log\hat{p}_{ijt} + (1-y_{ijt})\log(1-\hat{p}_{ijt})] + \lambda\|\mathbf{w}\|_2^2$$

Target metric and estimand:

$$\Delta_i = X_i^{\text{post}} - X_i^{\text{pre}}, \qquad \hat{\tau} = \bar{\Delta}_{\text{adaptive}} - \bar{\Delta}_{\text{traditional}}, \qquad H_0 : \tau = 0$$

7. Reproducibility

```
python -m pip install -r requirements.txt
python src/process_data.py && python src/visualize.py
jupyter nbconvert --to notebook --execute --inplace notebooks/01_data_analysis.ipynb
```

Tested from a fresh clone in a clean Python 3.11 environment: no errors, and the regenerated data and figures are byte-identical to the committed ones. Packages are pinned exactly; seed = 42 everywhere. `.gitignore` excludes `.ipynb_checkpoints/`, `__pycache__/`, `.venv/` and large raw dumps.